

**Definition 8.1**

The *direct product* of groups  $G_1, \dots, G_n$  is a group  $G_1 \times G_2 \times \dots \times G_n$  defined as follows:

- Elements:  $n$ -tuples  $(g_1, g_2, \dots, g_i)$  where  $g_i \in G_i$ .
- Group operation:

$$(g_1, g_2, \dots, g_n) \cdot (h_1, h_2, \dots, h_n) = (g_1 h_1, g_2 h_2, \dots, g_n h_n)$$

- The identity element:  $(e_1, e_2, \dots, e_n)$  where  $e_i$  is the identity element in  $G_i$ .
- Inverses:  $(g_1, g_2, \dots, g_n)^{-1} = (g_1^{-1}, g_2^{-1}, \dots, g_n^{-1})$ .

**Note.** We have:

$$|G_1 \times G_2 \times \dots \times G_n| = |G_1| \cdot |G_2| \cdot \dots \cdot |G_n|$$

### Theorem 8.2

The group  $G_1 \times \dots \times G_n$  is abelian if and only if each of the groups  $G_i$  is abelian.

### Recall:

- The *least common multiple* of integers  $n_1, n_2, \dots, n_k \geq 1$  is the smallest positive integer, denoted by  $\text{lcm}(n_1, \dots, n_k)$ , which is divisible by each of these numbers.
- If  $m > 0$  is an integer divisible by  $n_1, \dots, n_k$  then  $m$  is divisible by  $\text{lcm}(n_1, \dots, n_k)$ .

### Theorem 8.3

For  $i = 1, \dots, n$  let  $a_i \in G_i$ , and let  $(a_1, \dots, a_n) \in G_1 \times \dots \times G_n$ . Then

$$|(a_1, \dots, a_n)| = \text{lcm}(|a_1|, \dots, |a_n|)$$